Terra Imperium

Mathematics, pacing, fairness, and balance

Implementation research report • 29 October 2026

Prepared for sharing with Claude or a development team. Formulas are selectable text; citations are clickable. This document packages the recommendations from the preceding conversation, with notation formatted for implementation.

The highest-value fixes are one shared combat model, explicit pacing budgets, and AI campaigns that check supply before marching. More event cards will not fix turns that lack strategic choices.

Evidence boundary: sources establish mathematical models or documented game precedents. All proposed constants, target ranges, and Terra Imperium formulas are starting hypotheses, not published industry standards. Related mechanics in a shipped game do not imply that game uses the exact equation proposed here.

Game context: turn-based grand strategy on a real-Earth hex map, about 77 km per hex; 2000 BCE–2300 CE; 150 historical peoples; eight research-unlocked ages: Stone, Bronze, Classical, Kingdoms, Gunpowder, Modern, Information, Future. Cities, research, armies, diplomacy, and events operate on turns. Battles can be commanded in real time or auto-resolved. Tactical battles include squads, morale, walls, workers, and buildings. Target platform: phones in landscape.

Reported problems: forecasts disagree with battle results; too many empty turns and repetitive cards; Fast ages pass too quickly; first Settlers take around 28 turns; late gold has no purpose; AI wars are passive or surprising and armies idle or starve; naval battles are rare.

Deterministic implementation contract

Use a versioned PRNG with separate streams derived from (campaignSeed, subsystem, entityID, decisionID). Forecast sampling must never advance the live combat stream. Use fixed simulation ticks, stable entity iteration order, and fixed-point arithmetic or explicitly controlled rounding. Store simulation and ruleset versions with saves and replays.

Common notation: clamp(x,a,b) limits x to [a,b]. $\sigma(x) = 1 / (1 + \exp(-x))$. Unless stated otherwise, durations are Standard-speed turns. Fractional state should remain fractional internally even if the UI rounds it.

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Ranked implementation priorities

1. Combat resolution and auto-resolve

Problem: forecasts, Auto, and commanded battles disagree because they evaluate different versions of the fight.

Model and precedent

Lanchester's square law describes idealized aimed fire:

$$dA/dt = -\beta B;\ dB/dt = -\alpha A$$

Conserved quantity: $\alpha A^2 - \beta B^2$

A and B are surviving combatant counts. α and β are per-combatant casualty rates. Numerical superiority compounds because more survivors keep dealing more damage.

For contact-limited fighting, model engaged frontage:

$$dA/dt = -\beta E;\ dB/dt = -\alpha E;\ E = \min(A,B,W)$$

W is the maximum number of simultaneous duels. This creates a linear exchange relationship rather than square-law amplification.

These are analytical checks, not sufficient models for mixed armies attacking repairable cities. Research applied generalized Lanchester models to StarCraft combat prediction, including an exponent between linear and square laws. This does not establish that StarCraft's shipped engine uses Lanchester equations. StarCraft combat-prediction research

Make Auto a headless execution of the tactical simulation with a standard commander. A practical damage rule is:

$$D_{ij} = \Delta t \times n_i \times d_i \times q_{ij} \times c_{ij} \times m_i \times f_{ij} \times 100/(100+a_j)$$

D_{ij} is damage from squad i to target j during a tick. Δt is seconds per tick; n_i is living attackers able to engage; d_i is base damage per attacker per second; q_{ij} is hit probability or deterministic expected-hit factor; c_{ij} is the unit-counter multiplier; m_i is morale; f_{ij} is position; $a_j \geq 0$ is armor.

$$m_i = 0.75 + 0.25M_i;\ M_i \in [0,1]$$

Zero morale reduces outgoing damage 25%. It never cancels orders, routes squads, or disables movement. Model walls as real targets with HP, firing positions, and blocked paths. Workers repair with the same rates and resource rules in both modes. Victory must include objectives and timeouts, not only annihilation.

Honest forecasts

Run N samples through the same Auto simulator, varying legitimate uncertainty: seeded combat variation, plausible hidden forces, or documented deployment uncertainty.

$$\text{Estimated win probability } \hat{p} = w/N$$

w is objective victories. Report the Wilson interval from section 12. For 200 wins in 256 samples, show 78% Auto victory probability with approximately 73–83% statistical confidence interval. Also report casualty quantiles, such as "Typically lose 20–40%." A loss prediction interval is different from confidence in the win-rate estimate.

If all information and the live seed are fixed and known, the result is deterministic. Probability describes uncertainty over a defined ensemble. Commanded battles need a separate label: "Auto estimate: 78%. Command outcome depends on your tactics." Later, add skill-conditioned forecasts from observed player cohorts.

Starting constants, example, and tests

Start with fixed 0.1-second ticks; morale floor 0.75; 64 forecast samples initially and 256 on opening details. Cache by battle-state hash. Resolve simultaneous damage before removing casualties to avoid iteration-order advantage.

Worked example: with identical units, square-law combat predicts that 120 versus 100 leaves $\text{sqrt}(120^2-100^2) \approx 66$ attackers. Equal contact-limited exchanges leave 20. Frontage therefore changes results dramatically.

Headless tests: stratify by age, siege/open field, force ratio, composition, and terrain. Target calibration error below 5 percentage points in adequately populated bins. Baseline commanded-AI replay and Auto must match exactly with identical inputs.

Risk: narrow sampling intervals do not measure model error. Missing repairs or reinforcements can make forecasts confidently wrong.

2. Making commanded skill matter

Problem: players need positional advantages they can understand and execute on a phone.

Model and precedent

Use bounded positional bonuses:

$$f_{ij} = 1 + \min(0.35, 0.15F_{ij} + 0.10H_{ij} + 0.10X_j)$$

F_{ij} is 1 outside the target's frontal arc; H_{ij} is 1 for advantageous elevation; X_j is 1 when the target is engaged from sufficiently separated directions. Otherwise each is zero.

$$D_{covered} = D \times (1-c);\ 0 \le c \le 0.30$$

c is cover protection against ranged damage. Focus fire should help by removing enemy damage sources, not through an arbitrary stacking bonus. Preserve projectile travel and overkill.

A competent Auto commander can prioritize targets using:

$$V_j = d_j / (h_j/D_{assigned,j} + t_{travel,j} + \varepsilon)$$

V_j is priority; d_j is threatening target DPS; h_j is remaining target HP; $D_{assigned,j}$ is friendly DPS assigned to it; $t_{travel,j}$ is time to reach or hit; ε prevents division by zero. Evaluate only feasible targets and handle zero assigned DPS explicitly.

StarCraft combat-prediction research recognizes that control proficiency changes effective unit strength. Command policy belongs in the model; humans need no hidden damage bonus. Research paper

Starting constants and tuning

Flank +15%; elevation +10%; positional cap +35%; cover 20–30%. AI target reassessment every 0.5–1 second. Give Auto sensible focus fire and basic cover use. Expose facing arcs, flank arrows, and group commands; offer tactical pause if compatible with the design.

Worked example: 100 base DPS becomes 125 with flank and elevation. Against 20% cover it deals 100 DPS. Position cancels a defensive advantage without a tenfold swing.

Headless tests: scripted novice, baseline, and skilled controllers with equal armies and mirrored sides. Starting goal: skilled wins 60–70% of balanced encounters against baseline; Auto stays within roughly 10–20% of skilled survivors in favorable battles.

Risk: frame-perfect targeting and facing exploits reward tapping speed. Require sustained exposure, such as one second, before granting a flank bonus.

3. Economy, costs, inflation, and snowballing

Problem: early construction stalls while late income loses purchasing purpose.

Model and precedent

Price content in turns of representative output:

$$C_{a,k} = \tau_k \times P_a$$

$C_{a,k}$ is production cost for item k in age a; τ_k is desired build duration; P_a is benchmark city production measured offline. Never recalculate costs from a player's current output: better production must shorten construction.

$$P_i = p_0 + p_w L_i + B_i$$

$$Y_i = g_w L_i + T_i$$

P_i is production; L_i productive population; p_0 base production; p_w and g_w production and gold per productive population; B_i building production; T_i trade income.

$$G_{t+1} = G_t + Y_t - U_t - B_t$$

G is treasury; Y total gold income; U upkeep; B purchases and projects.

$$U_{admin} = uN + v \times \max(0, N-K)^2$$

N is city count; K administrative capacity; u base city upkeep; v the over-capacity coefficient.

Civilization VI scales several kinds of costs with speed. That is a precedent for coordinated schedules, not this exact turn-budget formula. Civ VI speed

Starting constants and tuning

First Settler: 8–12 turns. Basic improvement: 3–5. Core infrastructure: 6–10. Initial capacity K=4; research expands it. Illustrative upkeep: u=2 and v=1, scaling currency units by age.

Provide useful gold sinks: upgrades, construction acceleration, supply depots, ships, diplomacy, reconstruction, and public projects. For a repeatable project:

$$b(x) = b_{max} \times x/(x+k)$$

x is committed gold; b_{max} maximum benefit; k spending that buys half the maximum benefit. Projects compete for limited slots.

Worked example: an 8-production capital builds an 80-production Settler in ten turns. A later 40-production city builds a 240-production improvement in six turns.

Headless tests: treasury measured in turns of expenditure; worthwhile purchase availability; build duration; net income after sinks. Discretionary reserves commonly around 5–15 turns of expenses are a starting goal, not a forced band. Verify several worthwhile late-game purchases exist.

Risk: a stockpile alone is not inflation. Wealth-sensitive hidden price increases feel punitive. Prefer published costs, maintenance, and capacity limits. Catch-up should come from diffusion and rebuilding opportunities, not confiscating the leader's income.

4. Population and city growth

Problem: unbounded growth snowballs; unexplained hard stalls make food investment feel wasted.

Model and precedent

Use food-limited logistic growth and separate starvation:

$$S = F - cP$$

$$\Delta P = \min[rP \times \max(0,1-P/K), \max(0,S)/q] - \min[dP, \max(0,-S)/q]$$

$$P_{(t+1)} = \max(0,P_t+\Delta P)$$

P is population in consistent game units; F food supplied per turn; c consumption per population; K housing capacity; r maximum proportional growth rate; q food surplus per new population; d maximum starvation loss fraction per turn.

Logistic growth supplies the diminishing-growth shape; the food constraint is a design addition. Civilization VI also constrains food-driven growth through housing, using different thresholds. Civ VI housing

Starting constants and tuning

Start r=0.08, c=2, q=20, d=0.02. Keep fractional population internally. Housing overcapacity stops growth rather than killing people. Starvation requires actual food deficit; buffer food, initially around three turns of consumption.

Worked example: P=10, K=20, F=30 gives surplus 10. Logistic growth is 0.4; food permits 0.5. Actual growth is 0.4 population.

Headless tests: food and housing should have monotonic benefits. Aim for roughly 12–25 turns to double a small well-supported city, depending on proximity to capacity. Record time food-limited versus housing-limited.

Risk: rounding every turn freezes small cities. Prevent instant starvation from a single lost farm.

5. Research and age pacing

Problem: research costs fail to track science growth, so ages disappear before their content matters.

Model and interpretation

Interpret “square of its multiplier” as:

$$T_{a,s} = T_0 \times m_a^2 \times s$$

$T_{a,s}$ is target age duration; T_0 base duration; m_a age-duration multiplier; s speed duration factor, smaller for faster games.

If the intended requirement is to square the speed multiplier instead, use $T=T_0 \times m_a^2 \times s^2$. Fast $s=0.7$ then gives 49% duration, aggravating the reported problem unless the base duration rises.

Set the mandatory research-path budget to expected accumulated science:

$$C_{path,a,s} = \Sigma \text{ from } t=1 \text{ to } T_{a,s} \text{ of } S_{a,s}(t)$$

S is benchmark science output during that age and speed. Allocate this budget among mandatory technologies with weights summing to one. Optional technologies add time and must not accidentally shorten the age gate.

Diffusion:

$$d_j = \min(0.30, 0.30 f_j)$$

$$C_{effective,j} = C_{base,j} \times (1 - d_j)$$

f_j is the weighted fraction of contacted relevant civilizations that know technology j. Trade partners can receive greater weight than distant contacts.

Civilization VI discounts technologies behind the World Era and surcharges those ahead of it, a related catch-up precedent. Civ VI technology costs

Starting age schedule

With $T_0=20$, use these rounded durations (age: multiplier; Standard; Fast $s=0.75$):

Stone: 1.00; 20; 15.

Bronze: 1.10; 24; 18.

Classical: 1.20; 29; 22.

Kingdoms: 1.30; 34; 25.

Gunpowder: 1.35; 36; 27.

Modern: 1.40; 39; 29.

Information: 1.45; 42; 32.

Future: 1.50; 45; 34.

Total: about 269 Standard or 202 Fast turns. Future needs its own completion budget or explicit end condition.

Carry unused raw science forward. Apply modifiers to the destination technology’s cost; never multiply already modified overflow again.

Worked example: 36 turns averaging 30 science require 1,080 science on the mandatory path. Four equal gate technologies cost 270 each before diffusion.

Headless tests: median age duration within ±15% of target across benchmark policies. Verify units get at least one useful campaign before obsolescence. Test fastest research paths separately from average play.

Risk: science changes during an age, and speed changes economic development. Calibrate S separately by speed. Never raise costs dynamically because the player researched well.

6. Pacing, engagement, and event cards

Problem: frequent cards interrupt while the underlying game offers too few consequential choices.

Model and precedent

Track smoothed intensity:

$$I_{t+1} = 0.8I_t + 0.2X_t$$

I is tension; X in [0,1] combines recent losses, threatened cities, shortages, and unresolved commitments. Use a state machine: build-up → pressure → peak → recovery.

Left 4 Dead's director uses intensity and pacing phases. This is a documented shipped precedent, but shooter timings should not transfer directly to strategy. Michael Booth: The AI Systems of Left 4 Dead

For discretionary event timing:

$$p(t)=0 \text{ when } t<c; \text{ otherwise } p(t)=1-\exp(-\lambda)$$

t is turns since the last event; c cooldown; λ the post-cooldown hazard.

Choose among eligible cards:

$$w_i = b_i \times r_i \times \exp(-n_i)$$

b_i is base weight; r_i in [0,1] contextual relevance; n_i recent appearances. Hard-exclude recently used IDs and families. If the pool is empty, show no card.

Measure opportunities rather than clicks:

$$D_t = \Sigma_j \text{ indicator(choice j has two viable alternatives and persistent consequences)}$$

D_t is meaningful decision opportunities. Approximate viability through policy rollouts and validate with humans.

Starting constants and tuning

Global card cooldown 3 turns; λ=0.35, giving about three additional eligible turns on average. No identical card for 30 turns or same family for 10. Major negative events 8–12 turns apart. After severe losses, allow 3–5 turns without director-added punishment.

Worked example: I=0.7 and X=0.1 gives next intensity 0.58. Allow recovery rather than immediately replacing a lost army with another crisis.

Headless tests: initially aim for median 2–4 meaningful decisions per turn and fewer than 15% of turns with none. Enforce zero cooldown violations. Measure late game separately. Agents cannot establish whether a decision feels meaningful.

Risk: players can manipulate a director. Schedule optional opportunities and complications, not fabricated enemy armies or invalidated success. Late projects, diplomatic blocs, and competing commitments must supply the actual choices.

7. AI wars, utility, gathering, and supply

Problem: AI lacks persistent, feasible campaign plans.

Model and precedent

Score proposed wars:

$$U_{war} = pV - (1-p)L - C_{march} - C_{occupation} + gG$$

p is estimated campaign success; V achievable objective value; L defeat loss; C_march and C_occupation logistics and holding costs; G in [0,1] grudge; g maximum grudge contribution. All value terms use common utility units.

Estimate p from deployable relative strength, alliances, enemy commitments, and supply—not total empire army size. For feasible actions:

$$U(a) = \Sigma_k w_k \times x_k(a)$$

x_k are normalized considerations and w_k weights. No viable route, binding truce, or insufficient supplies are hard gates, not weak penalties.

Utility curves are an established game-AI technique. EU4 provides a shipped precedent for visible hostility, strategic interests, and war preparation. GDC: Improving AI Decision Modeling Through Utility Theory • EU4 diplomatic-feedback documentation

Supply after carried reserves run out:

$$\delta = \max(0,(D-S)/D);\ a = \min(0.08,0.10\delta)$$

D>0 is supply demand; S available supply; δ shortage fraction; a attrition per turn.

Starting constants and tuning

Gather to 70% forecast victory probability while retaining home defense. Carry three march-turns of supplies. Replan after three turns without objective progress. Require a new action to exceed the present plan’s utility by 15% before switching. Telegraph detectable mobilization 2–3 turns ahead.

Worked example: p=.75, V=100, L=60, logistics plus occupation=25 gives 75−15−25=35 utility. Demand 100 with supply 60 causes 4% attrition per turn after reserves expire.

Headless tests: over 90% of campaign departures have viable projected routes; fewer than 10% arrive below 80% departure strength solely from preventable attrition. Track idle armies without assigned strategic roles.

Risk: excessive caution creates permanent peace. Impose planning deadlines and allow smaller objectives—raids, border forts, concessions—when full invasion is infeasible.

8. Diplomacy, opinion, trust, and tribute

Problem: diplomacy is opaque and repeated offers exploit random acceptance.

Model and precedent

Separate temporary opinion and durable trust:

$$O_t = O_{structural} + \Sigma_e v_e \times 2^{(-(t-t_e)/h_e)}$$

$O_{structural}$ is ongoing alignment/conflict; v_e an event's opinion effect; t_e occurrence turn; h_e half-life.

$$T_{(t+1)} = clamp(T_t + \eta H_t - \rho B_t, 0, 1)$$

H_t measures honored meaningful commitments; B_t betrayal; η and ρ update strengths. EU4 distinguishes relations, attitudes, trust, and favors, and uses promises of territory in war diplomacy. EU4 documentation

Offer acceptance:

$$p_{accept} = \sigma(\Delta U/\tau)$$

ΔU is recipient utility gain versus its best alternative; τ measures modeled uncertainty or personality variation. If acceptance is fully exposed and deterministic, show Accepts or Rejects. Show probability only when uncertainty genuinely exists.

Tribute present value:

$$PV = q \times [1-(1+r)^{(-H)}] / r$$

q is payment per turn; H duration; r discount rate. Subtract default risk and strategic costs in utility calculations.

Starting constants and tuning

Minor insult half-life 10 turns; serious aggression 40. Honored substantial treaty: trust +0.03. Betrayal: −0.25. Tribute discount r=.03.

Worked example: 10 gold per turn for ten turns has PV≈85.3. $\Delta U=20$ and $\tau=15$ gives 79% acceptance probability.

Headless tests: shown probability calibrated within 5 percentage points. Repeated identical offers, subdivided gifts, and treaty loops must not create trust or new acceptance rolls.

Risk: lock uncertainty to negotiation ID and meaningful world state. Editing an offer by one gold must not reroll personality.

9. Naval encounters, piracy, and blockades

Problem: fleets have little reason to occupy the same water.

Model and precedent

Create valuable shipping lanes, then exposure-based encounters:

$$p_{encounter} = 1-\exp(-\lambda E)$$

λ is encounter rate per exposure unit. E combines overlapping routes, duration, detection, and actual hostile presence. No hostile fleet or pirates means E=0.

Piracy:

$$L_{piracy} = V \times \ell_{max} \times P/(P+E_s+k)$$

V is route value; P pirate pressure; E_s escort strength; k>0 baseline resistance; ℓ_{max} maximum loss fraction.

Blockade:

$$b = B/(B+D+k_b);\ L_{blockade} = b \times V_{sea}$$

B is blockading power; D defending naval/coastal power; k_b port resistance; V_sea exposed maritime income.

EU4 offers trade protection, privateering, pirate hunting, fleet hunting, and blockade missions. Official Mare Nostrum developer diary

Starting constants and tuning

On actively contested routes, target encounter probability 0.15–0.25 per exposed turn. Established unprotected piracy: 5–15% expected trade loss. A substantial blockade should threaten enough income or imported supply to justify a response. Automated escort and patrol orders are essential on phones.

Worked example: λE=.2 gives 18% encounter probability, about one per 5.5 exposed turns at a constant independent rate. B=60, D=20, k_b=20 means a blockade removes 60% of exposed maritime income.

Headless tests: measure per contested route-turn, not per campaign. Aim for one meaningful interception per 4–8 exposed turns. Verify AI responds to profitable blockade targets.

Risk: arbitrary random sea battles undermine readability. Use actual geometry and forces, with intelligence indicators showing risk.

10. Map, settlement, and expansion

Problem: weak site valuation produces city spam, empty regions, and implausible expansion.

Model and precedent

Use discounted marginal site value:

$$V(s) = \Sigma \text{ from } t=1 \text{ to } H \text{ of } \gamma^{(t-1)}[Y_s(t)-U_s(t)-R_s(t)] - C_s$$

s is candidate site; H horizon; γ discount factor; Y projected marginal yield in common utility units; U upkeep/logistics; R expected defense/loss cost; C founding opportunity cost. Do not count shared workable land twice.

Aggressive expansion:

$$AE_{ij,t+1} = 2^{(-1/h)}AE_{ij,t} + \Sigma_c k v_c \exp(-d_{jc}/\ell)(1+\rho a_{jc})$$

AE_{ij} is observer j's alarm about conqueror i; h half-life; c newly annexed city; v_c city value; d_{jc} observer distance from conquest; ℓ geographic sensitivity; a_{jc} in [0,1] affinity with victim; k and ρ scale and affinity weights.

EU4 supplies the related aggressive-expansion and coalition precedent. This distance-decay equation is a proposal. EU4 relations and aggressive expansion

Starting constants and tuning

City centers at least three hex steps apart—roughly 231 km if 77 km means center spacing. Test preferred separation of 3–5 hexes. Use H=20, γ=.95, AE half-life 20 turns, geographic scale ℓ=8 hexes.

Treat cities as regional centers: this resolution cannot represent every historical city independently.

Worked example: net utility 8/turn for 20 turns is about 8×12.83=103. Founding cost 60 leaves net site value 43.

Headless tests: settlement density, intercity travel time, contested frontier frequency, starting opportunity, and identical-policy comparisons across all 150 starts.

Risk: blanket spacing makes small islands and narrow peninsulas unusable. Add explicit geographic exceptions, not hidden AI exemptions.

11. Legacy scoring and victory

Problem: final army size or territory erases earlier achievements and favors certain starts.

Model and precedent

Use permanent bounded achievements:

$$L_i = 1000 \times \Sigma_a v_a \times \Sigma_k w_k[1-\exp(-x_{iak}/b_{ak})]$$

L_i is legacy; a ranges across eight ages; v_a are age weights summing to one; k ranges across K categories; w_k category weights sum to one; x_{iak} is qualified achievement; b_{ak} is its published benchmark.

Suggested categories: prosperity, science, culture, diplomacy, military/security. Reward durable outcomes and unique achievements, not raw spending or repeated recaptures. Humankind uses accumulated Fame and achievements across eras to determine victory. Amplitude: Shaping Your Legacy

Starting constants and tuning

Equal ages $v_a=1/8$; five equal categories $w_k=.2$. One benchmark earns 63% of a component; two benchmarks earn 86%. Preserve achievements after elimination. A longer age must not generate unlimited additional score.

Worked example: $x=b$ in science gives $1000\times1/8\times.2\times.632\approx15.8$ legacy points for that age.

Absolute achievement and performance relative to a difficult start are different concepts of fairness. Keep public legacy comparable, with an optional separate start-adjusted rating:

$$Z_i = (L_i-\mu_{start,i}) / \max(\sigma_{start,i},\sigma_{min})$$

μ_{start} and σ_{start} are frozen benchmark mean and standard deviation for the starting position; σ_{min} prevents unstable division.

Headless tests: compare strategy archetypes and starts. Investigate categories differing by over 10% in attainable score under similarly capable policies. Verify peaceful and military approaches can both win.

Risk: per-capita metrics alone reward staying tiny; absolute totals alone favor rich starts. Do not hide moving goalposts in main scoring. Freeze benchmarks before campaigns.

12. Balance testing and sample sizes

Problem: small correlated test sets make noise look like improvement.

Confidence intervals

For observed proportion $\hat{p}=w/n$, use the 95% Wilson interval:

$$c = [\hat{p}+z^2/(2n)] / [1+z^2/n]$$

$$h = z \times \mathrm{sqrt}[\hat{p}(1-\hat{p})/n + z^2/(4n^2)] / [1+z^2/n]$$

$$\text{Interval} = [c-h,c+h];\ z=1.96$$

n is independent trials; w successes. [NIST confidence-interval guidance]

Approximate planning near 50%:

$$n \approx 1.96^2 \times 0.25/e^2$$

e is desired interval half-width. Required independent samples: ±10 percentage points ≈97; ±5 ≈385; ±3 ≈1,068; ±2 ≈2,401.

Precision differs from power. Detecting a change from 50% to 55% with 80% power at two-sided 5% significance needs about 1,565 independent observations per version, before multiple-comparison adjustments.

Practical workflow

1. Run 100 seeds for smoke tests and catastrophic failures.
2. Use 400–1,000 paired campaign seeds for directional balance evaluation.
3. Use power-based confirmation for consequential changes and small effects.
4. Rotate starts, peoples, opponent policies, and player order.
5. Keep a held-out seed set.

Run old and new versions on the same seeds. For continuous metrics, analyze per-seed differences using intervals or campaign-level bootstrap. For paired binary outcomes, use a paired method such as McNemar’s test.

A campaign is a cluster: thousands of its turns or battles are not thousands of independent campaign samples. The 150 nations in one world also share conditions. Valve documented accelerated automated testing with survivor bots in Left 4 Dead; this supports agent-driven simulation, not any specific sample count. [Booth presentation]

Runaway and collapse metrics

$$s_i(t)=P_i(t)/\Sigma_j P_j(t);\ H(t)=\Sigma_i\, s_i(t)^2$$

P_i is a fixed weighted power index; s_i is share; H concentration. Compute separate economy, military, and score versions to avoid hiding imbalance.

Track the probability that the leader at 25%, 50%, and 75% campaign progress wins; leadership changes; bottom-quartile recovery into the top half; collapse defined as population or productive capacity falling over 40% within ten turns; treasury, starvation, inactivity, age duration, and calibration.

Worked example: four equal powers give H=.25. Shares .7,.1,.1,.1 give H=.52.

Starting triggers: investigate when the halfway leader wins over 85% of equal-policy campaigns or unintended peacetime collapse exceeds 5% of nations. These are investigation triggers, not mandatory laws.

Risk: 150 peoples produce many chance “significant” results. Predeclare primary metrics, correct multiplicity for confirmatory comparisons, and inspect effect sizes. Simulation establishes mechanics; human sessions establish readability and enjoyment.

Ranked implementation priorities

The ten changes most likely to improve fun per hour of work, assuming accessible combat, economy, event, and AI systems.

1. Label forecasts honestly and use Auto’s actual model

Restores trust immediately. Begin with accurate Auto labeling while shared tactical execution work proceeds. Shared combat execution may require substantial work, but addresses the most damaging trust failure.

2. Cut first Settler time to 8–12 turns

Opens exploration, competition, and city specialization much earlier.

3. Add global event cooldowns and family exclusions

Small implementation effort directly addresses repetitive interruptions.

4. Set research costs from age-duration budgets

Gives each age time to produce strategic consequences, especially on Fast.

5. Require supply-feasible AI departures

Removes conspicuously broken invasions and creates meaningful logistics targets.

6. Add campaign objectives, progress checks, and replanning deadlines

Converts idle armies into understandable pressure.

7. Add useful recurring gold expenditures

Makes late income support upgrades, fleets, reconstruction, and competing projects.

8. Expose mobilization and diplomatic reasons

Makes wars feel caused by the world and gives players time to respond.

9. Add bounded visible flank and cover advantages

Rewards tactical judgment with rules readable on a small screen.

10. Make shipping lanes valuable and assign naval missions

Creates reasons to contest the sea, a stronger foundation than increasing random encounter rates.